Brief lookback cues content generation in spontaneous multi-sentence text production

Multi-sentence text composition is typically remarkably fluent. Even at major planning units—at the start of sentences, for example—competent writers often proceed with negligible hesitation [1]. This suggests mostly parallel, just-in-time [2] processing, with message preparation occurring concurrently with output, and meaning generated as needed during production. Pausing, when it does occur, is often associated with eye movement back into the writer's own text. This is typically characterized by writers hopping between informationally rich words, rather than sustained reading, and often occurs without error correction [3]. It seems probable that this lookback serves, in part, to reinstate a representation of the message of the text already written, and therefore to cue what to say next, thus maintaining thematic coherence and production fluency. This hypothesis has not previously received a direct test.

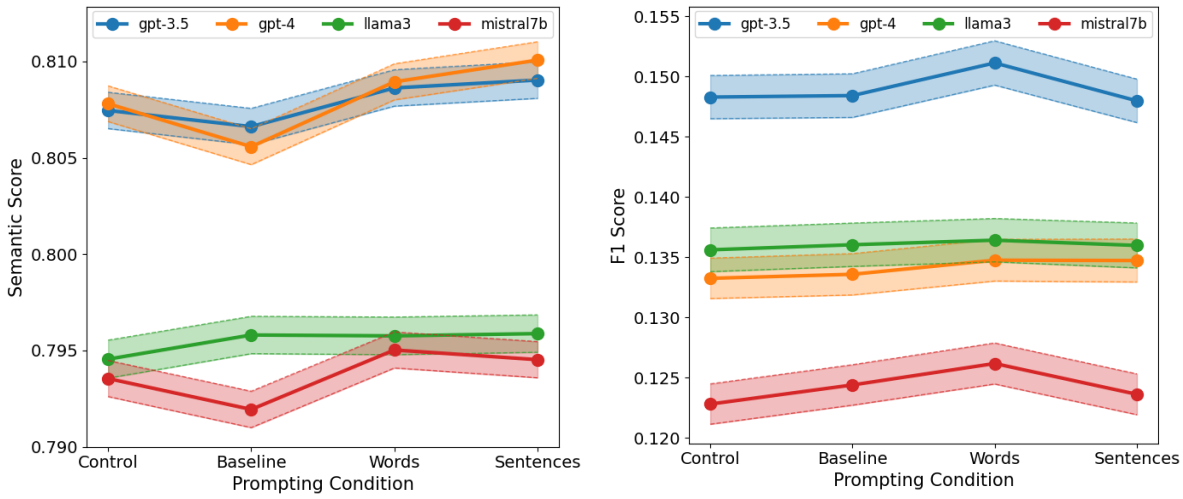
In this study, we logged keystrokes and tracked eye movement in 30 competent adult writers each composing two short argumentative essays, using methods that permitted extraction of the words and sentences fixated within the writer's own, growing text [4]. We identified where writers paused and looked back into their text fixating a minimum of three words, and then continued writing without any intervening error correction or revision. For each of these lookback/completion events we compared the writer's completion (the text they wrote next up to the sentence terminator) with completions produced by a generative LLM under four different prompting conditions. In all conditions, the LLM was prompted with the essay topic and with the writer's existing text from the beginning of the current paragraph up to the cursor location at the time of the pause. To this prompt, we added one of the following: open-class words that the writer fixated ("words" condition); the text of the sentences that contained fixated words ("sentences" condition); a matched number of open-class words that were not fixated ("control" condition); and nothing ("baseline" condition). The "sentences" condition was included on the grounds that fixating one or more open-class words within a recently-written sentence is likely to activate the meaning of the sentence as a whole.

We sampled 10 LLM responses per lookback/completion event, with each of 4 different language models (GPT-3.5, GPT-4, LLaMa3-8B and Mistral7B). *Semantic similarity* between human and LLM completions was established by embedding using OpenAI's text-embedding-ada-002 model and computing cosine similarity. We also report word-token-based *Jaccard* and *F1* (the harmonic mean of precision and recall) indices to quantify Surface similarity.

Frequentist linear mixed effects models with maximal random effects structure and prompting condition as fixed effect gave significantly better fit than an intercept-only model for all three similarity measures. Planned comparisons showed significantly greater semantic similarity between LLM and human completions when LLM prompts included fixated words or sentences. Adding fixated words (but not sentences) also improved surface-level similarity. Effect estimates are given in Figure 1.

Semantic similarity between human and LLM text was greater when words fixated by the human writer immediately prior to production were allowed to shape the LLM response. To our knowledge, this offers the first tentative evidence that lookback during text production supports rapid, inline content planning.

Figure 1. Estimated mean similarity between human-writer sentence completions and completions from LLMs prompted without and with information about the writer's fixations within their own text, with 95% CI.



(a) Semantic similarity (b) Surface similarity (F1)

References

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